UNIT - IV

Slope-Deflection Method

1. Introduction

- The **slope-deflection method** was introduced by **G. A. Maney (1915)**. It is a **displacement (stiffness) method**: the **joint rotations and joint translations** are the unknowns (kinematic indeterminacy D_k).
- The end moments of every member are written in terms of the **joint displacements** (slope θ and chord rotation ψ) and the loads. Then **joint equilibrium** equations give the unknown displacements.
- It is suitable for **continuous beams and rigid frames**, especially when $D_k < D_s$.

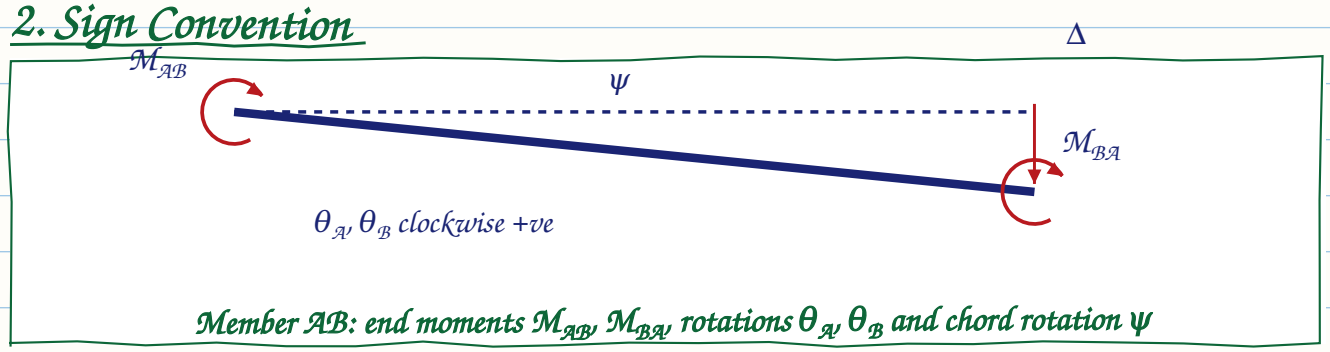
Main idea

Member end moment = (fixed end moment due to loads) + (moment due to end rotations and relative displacement).

Assumptions

1. Material is linearly elastic; small deflections; E and I constant for each member.
2. Joints are **rigid** — all members meeting at a joint rotate through the same angle.
3. **Axial and shear deformations are neglected**; only bending deformation considered.
4. Distortion due to axial force is neglected, so the **length of members remains unchanged**.
5. Supports unyielding unless settlement is stated.

2. Sign Convention



Member AB: end moments M_{AB}, M_{BA}, rotations θ_A, θ_B and chord rotation ψ

- **Clockwise** end moments (acting on the member), **clockwise** joint rotations θ_A, θ_B, and **clockwise chord rotation ψ** are all **positive**.
- $\psi = \Delta / L$, where Δ is the relative transverse displacement of the ends; it is **positive (clockwise)** when the far end **B moves downward** relative to A (or A rises relative to B).
- At a joint the moment of the member on the joint is opposite to the moment of the joint on the member (**Newton's third law**): ΣM at a joint = 0.
- For BMD: a clockwise moment at the **left** end of a member or an anticlockwise moment at the

right end gives **sagging** .

3. Derivation of Slope-Deflection Equations

- Take a member AB of length L, constant EI, with end moments M_{AB} and M_{BA}, end rotations θ_A, θ_B and chord rotation ψ. Treat it as a simply supported beam of span L carrying end moments (**M/EI diagram = two triangles**) — use the **moment-area / conjugate beam** method.
- The rotation of each end relative to the chord ($\theta - \psi$):

From conjugate beam (end moments only)

$$\theta_A - \psi = (L/6EI)(2M_{AB} - M_{BA})$$
$$\theta_B - \psi = (L/6EI)(2M_{BA} - M_{AB})$$

- Check: M_{AB} alone $\rightarrow \theta_A = ML/3EI$ (clockwise) and the far end rotates $ML/6EI$ in the opposite sense.
- Solve for M_{AB} and M_{BA}: multiply the first by 2 and add the second:
- $12EI(\theta_A - \psi)/L + 6EI(\theta_B - \psi)/L = 3M_{AB} \rightarrow$ **$M_{AB} = (2EI/L)(2\theta_A + \theta_B - 3\psi)$** ; similarly M_{BA}.

Slope-deflection equations (with loads)

$$M_{AB} = F_{AB} + (2EI/L)(2\theta_A + \theta_B - 3\psi)$$
$$M_{BA} = F_{BA} + (2EI/L)(2\theta_B + \theta_A - 3\psi)$$

- F_{AB}, F_{BA} = **fixed end moments** due to the loads on the span (clockwise +ve). They are found by assuming both ends fixed (Unit III).
- Using **$k = I/L$** (relative stiffness): $M_{AB} = F_{AB} + 2EK(2\theta_A + \theta_B - 3\psi)$.

Physical meaning of terms

Term	Meaning
$4EI\theta_A/L$	moment needed to rotate end A by θ_A with B fixed — **stiffness $4EI/L$**
$2EI\theta_B/L$	**carry-over** moment: half of $4EI\theta/L$ goes to the far end (carry-over factor ½)
$-6EI\psi/L$	moment due to relative displacement of the ends
F_{AB}	fixed end moment due to loads

4. Fixed End Moments (clockwise positive)

Loading on span AB (length L)	F_{AB}	F_{BA}
UDL w over whole span	$-wL^2/12$	$+wL^2/12$
Central point load P	$-PL/8$	$+PL/8$
Point load P at a from A (b from B)	$-Pab^2/L^2$	$+Pa^2b/L^2$

Loading on span AB (length L)	F_{AB}	F_{BA}
Two equal loads at L/3 and 2L/3	$-2PL/9$	$+2PL/9$
UVL zero at A, w at B	$-wL^2/30$	$+wL^2/20$
UVL w at A, zero at B	$-wL^2/20$	$+wL^2/30$
Couple M_0 at a from A (clockwise)	$+M_0\,b(2a-b)/L^2$	$+M_0\,a(2b-a)/L^2$
Settlement Δ of B (downward), no load	$-6EI\Delta/L^2$	$-6EI\Delta/L^2$

→ Remember: at the **left end the FEM of a downward load is anticlockwise (–)** and at the **right end clockwise (+)**. Equal magnitudes for symmetrical loads.

5. Modified Equation — Far End Hinged or Roller

- If the far end B of the member is **hinged or a roller** ($M_{BA} = 0$), substitute to remove θ_B:

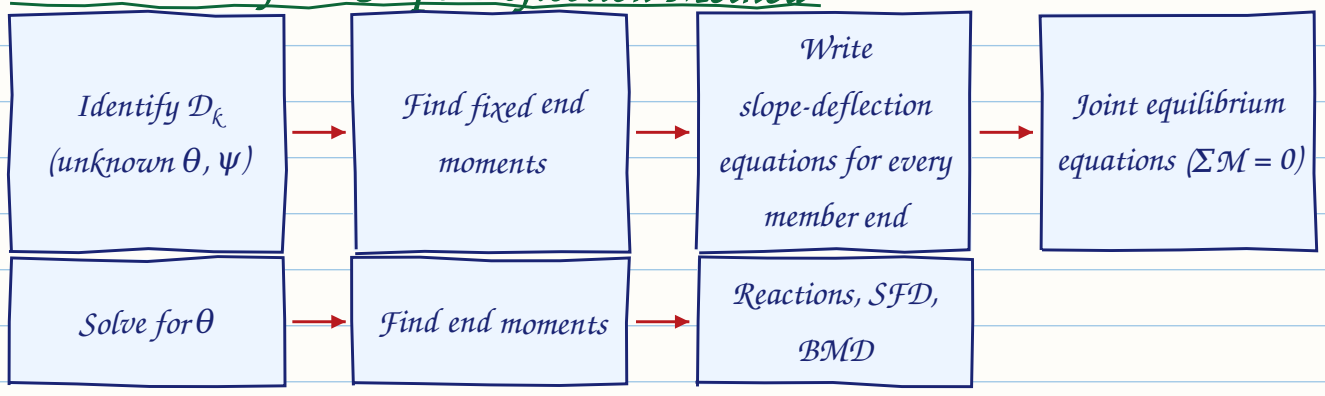
Modified slope-deflection equation

$$M_{AB} = F'_{AB} + (3EI/L)(\theta_A - \psi) \text{ where } F'_{AB} = F_{AB} - \tfrac{1}{2} F_{BA}$$

- Stiffness of the member becomes **3EI/L** (instead of 4EI/L); no carry-over to the hinged end.
- Modified fixed end moments: **UDL: $-wL^2/8$** ; central point load: **$-3PL/16$** .
- It reduces the number of unknowns — use it for the **end spans simply supported** .

Far end condition	Stiffness	Carry-over
Fixed	4EI/L	1/2
Hinged / roller	3EI/L	0
Free (cantilever)	0 (no stiffness)	—

6. Procedure of the Slope-Deflection Method



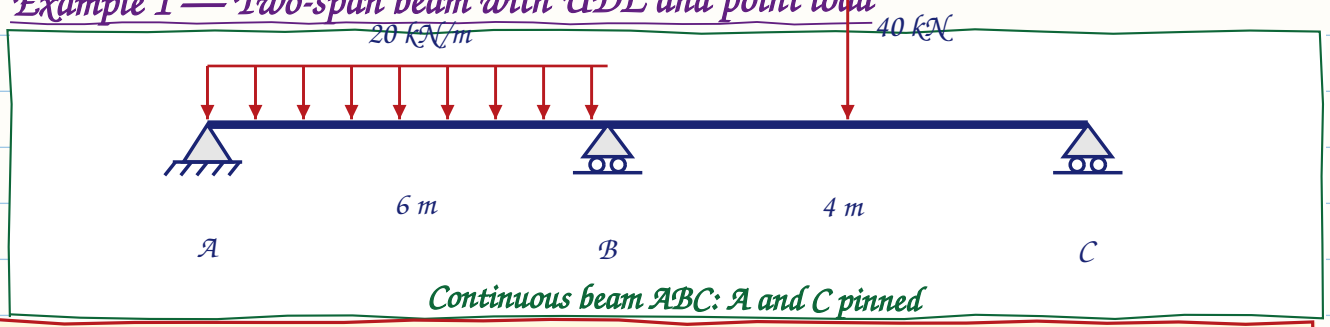
1. **Number of unknowns:** all joint rotations that are not zero (fixed ends have $\theta = 0$) and sway displacements.
2. Find F_{AB}, F_{BA} for loaded spans, ψ for members whose ends have relative displacement (settlement or sway).
3. Write the **two equations for each member** (or modified one if the far end is hinged).

4. Write **equilibrium equations** : at each joint $\Sigma M = 0$ (algebraic sum of the member end moments = 0, or = applied external moment). For a free-end overhang, the moment is known.
5. For frames with sway, add the **shear (storey) equation** ; in this unit sway is excluded.
6. Solve simultaneous equations ($EI\theta$ will come out); back-substitute for the end moments (check ΣM at joints = 0).
7. Find **reactions** : R = (free reaction) $\pm (M_{BA} + M_{AB})/L$; draw **SFD and BMD** .

→ Tip: Always work with $EI\theta$ as the unknown. Keep the units consistent (kN, m).

7. Continuous Beams without Settlement — Solved Examples

Example 1 — Two-span beam with UDL and point load



Continuous beam ABC: A and C pinned

Problem

AB = 6 m carries UDL 20 kN/m; BC = 4 m carries a 40 kN load at its centre. EI is constant. Ends A and C are simply supported. Find the moments and reactions.

- Only unknown: θ_B (A and C hinged → use the modified equation).
- FEMs: $F_{BA} = +wL^2/12 = +60$, $F_{AB} = -60 \rightarrow F'_{BA} = 60 + 30 = +90$. For BC: $F_{BC} = -PL/8 = -20$, $F_{CB} = +20 \rightarrow F'_{BC} = -20 - 10 = -30$.
- **$M_{BA} = 90 + (3EI/6)\theta_B = 90 + 0.5\,EI\theta_B$; $M_{BC} = -30 + (3EI/4)\theta_B = -30 + 0.75\,EI\theta_B$**
- Joint B: $M_{BA} + M_{BC} = 0 \rightarrow 60 + 1.25\,EI\theta_B = 0 \rightarrow$ **$EI\theta_B = -48$** .
- $M_{BA} = 90 - 24 =$ **$+66$ kN·m** ; $M_{BC} = -30 - 36 =$ **-66 kN·m** . So **$M_B = 66$ kN·m (hogging)** ; $M_A = M_C = 0$.
- Reactions: $R_A = 60 - 66/6 =$ **49 kN** ; $R_B = (60 + 11) + (20 + 16.5) =$ **107.5 kN** ; $R_C = 20 - 66/4 =$ **3.5 kN** ; check: 49 + 107.5 + 3.5 = 160 = 120 + 40 √.
- Check by three-moment theorem: $2M_B(10) = 1080 + 240 \rightarrow M_B = 66$ √.
- Sagging in AB: $x = 49/20 = 2.45$ m from A → $M = 49 \times 2.45 - 10 \times 2.45^2 =$ **60.0 kN·m** . Under the 40 kN load (2 m from C): $M = 3.5 \times 2 =$ **7 kN·m** (small, since the hogging moment at B dominates).

Example 2 — Beam with a fixed end

Problem

ABC with A fixed, B and C rollers, AB = BC = 6 m, UDL 20 kN/m on both spans.

- $\theta_A = 0$; unknown θ_B (C roller → modified for BC). $F_{AB} = -60$, $F_{BA} = +60$, $F'_{BC} = -60 - 30 =$

-90.

- $M_{AB} = -60 + (2EI/6)(\theta_B) = -60 + EI\theta_B/3$
- $M_{BA} = +60 + (2EI/6)(2\theta_B) = 60 + 2EI\theta_B/3$
- $M_{BC} = -90 + (3EI/6)\theta_B = -90 + 0.5\,EI\theta_B$
- Joint B: $M_{BA} + M_{BC} = 0 \rightarrow -30 + 1.1667\,EI\theta_B = 0 \rightarrow$ **$EI\theta_B = 25.71$** .

Results

$M_{AB} = -51.43$, $M_{BA} = +77.14$, $M_{BC} = -77.14$ kN·m (matches the three-moment result of Unit III)

- Span AB: $R_A = 60 + (51.43 - 77.14)/6 =$ **55.71 kN**, R_B(left) = 64.29 kN. Span BC: $R_C = 60 - 77.14/6 =$ **47.14 kN**, R_B(right) = 72.86 kN.
- **$R_B = 137.14$ kN**; check: 55.71 + 137.14 + 47.14 = 240 = total load √.

Example 3 — Three-span continuous beam (equal spans, UDL)

- Spans L, UDL w, ends simply supported; symmetry: $\theta_B = -\theta_C$. Using modified equation for the end spans and full equation for the middle span:
- $M_{BA} = wL^2/8 + (3EI/L)\theta_B$; $M_{BC} = -wL^2/12 + (2EI/L)(2\theta_B + \theta_C) = -wL^2/12 + (2EI/L)\theta_B$.
- $\Sigma M_B = 0$: $wL^2/8 - wL^2/12 + (5EI/L)\theta_B = 0 \rightarrow EI\theta_B = -wL^3/120$.
- **$M_{BA} = wL^2/8 - wL^2/40 = wL^2/10$** √ (matches three-moment result).

Example 4 — Beam with overhang

- A cantilever overhang CD: its end moment at C is fixed: **$M_{CD} = -$ (load × arm)** ; write only M_{CB} equation with this known moment.
- Joint C equation: $M_{CB} + M_{CD} = 0 \rightarrow M_{CB} = +$ load × arm.

8. Continuous Beams with Settlement of Supports

- A support settlement Δ produces a **chord rotation** $\psi = \Delta / L$ in the adjacent spans.

Case	ψ
Span AB: B lower than A by Δ	$\psi_{AB} = +\Delta / L$ (clockwise)
Span BC: C higher than B (B lower)	$\psi_{BC} = -\Delta / L$ (anticlockwise)
Both ends settle equally	$\psi = 0$

- ψ is **the same for both ends** of a member.

Example 5 — Settlement of the middle support

Problem

Two equal spans $AB = BC = 6$ m, UDL 20 kN/m, $EI = 30000$ kN·m². A and C pinned. Support B settles by 10 mm. Find the support moment.

- $\psi_{AB} = +0.01/6 = 0.001667$; $\psi_{BC} = -0.001667$.
- $M_{BA} = F'_{BA} + (3EI/L)(\theta_B - \psi_{AB}) = 90 + 15000\,(\theta_B - 0.001667)$
- $M_{BC} = F'_{BC} + (3EI/L)(\theta_B - \psi_{BC}) = -90 + 15000\,(\theta_B + 0.001667)$
- Joint B: $30000\,\theta_B = 0 \rightarrow \theta_B = 0 \rightarrow M_{BA} = 90 - 25 =$ **65 kN·m** ; $M_{BC} = -90 + 25 =$ **– 65 kN·m** .
- Without settlement 90 kN·m; **settlement reduces the support moment** by $3EI\Delta/L^2 = 25$ kN·m.
- $R_A = 60 - 65/6 =$ **49.17 kN**, $R_C =$ **49.17 kN**, $R_B = 240 - 98.33 =$ **141.67 kN**.

Example 6 — Settlement of one end support

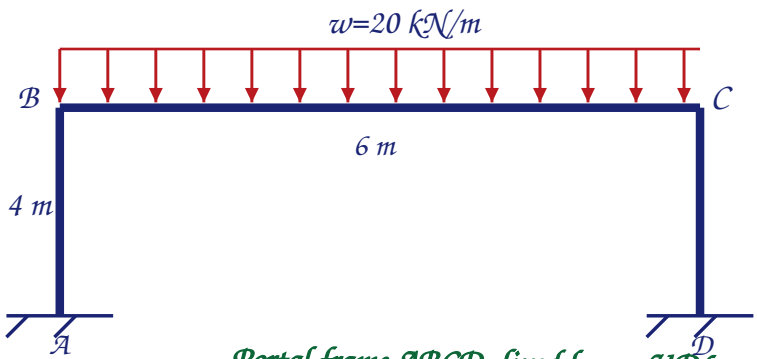
- If the end support C settles by Δ (downward), only span BC gets a chord rotation: C is lower than B, so the chord rotates clockwise and **$\psi_{BC} = +\Delta / L$** . Solve with θ_B as the unknown in the same way.

→ Always decide the sign of ψ with a quick sketch: **the member rotates clockwise** → ψ positive.

9. Single-Bay Portal Frames without Sway

- A portal frame is **without sway** if the frame and its loading are **symmetrical** or if the lateral movement of the beam is **prevented** by a support.
- Then the chord rotation $\psi = 0$ for **all** members (columns do not sway, beam has no differential movement).
- **Joint rotations θ_B and θ_C are the unknowns** (fixed bases: $\theta_A = \theta_D = 0$); for hinged bases use the modified equation for the columns.
- For symmetrical loading: **$\theta_C = -\theta_B$** — only one unknown.

Example 7 — Fixed-base portal frame with UDL on the beam



Portal frame ABCD, fixed bases, UDL on beam BC

Problem

Columns AB and DC are 4 m high; beam BC = 6 m; UDL 20 kN/m on BC; EI is the same for all members. A and D fixed.

- Symmetry → $\theta_C = -\theta_B$. FEMs for BC: $F_{BC} = -60$, $F_{CB} = +60$.
- $M_{AB} = (2EI/4)(\theta_B) = 0.5\,EI\theta_B$; $M_{BA} = (2EI/4)(2\theta_B) = EI\theta_B$
- $M_{BC} = -60 + (2EI/6)(2\theta_B + \theta_C) = -60 + (EI/3)\,\theta_B$
- Joint B: $M_{BA} + M_{BC} = 0 \rightarrow EI\theta_B\,(1 + 1/3) = 60 \rightarrow$ **$EI\theta_B = 45$** .

End moments

$M_{AB} = 22.5$ kN·m, $M_{BA} = 45$, $M_{BC} = -45$, $M_{CB} = +45$, $M_{CD} = -45$, $M_{DC} = -22.5$ kN·m

- Horizontal reactions: $H_A = H_D = (M_{AB} + M_{BA})/h = 67.5/4 =$ **16.88 kN** (acting inwards at the bases). Vertical: $V_A = V_D = wL/2 =$ **60 kN**.
- BMD: beam hogging **45 kN·m** at each end; centre sagging = $wL^2/8 - 45 = 90 - 45 =$ **+ 45 kN·m** ; columns: 22.5 at the base and 45 at the top.
- Check: $H_A + H_D$ horizontal forces balance (equal and opposite); $\Sigma V = 120$ = total load √.

Example 8 — Beam stiffer than columns ($I_{beam} = 2\,I_{column}$)

- $M_{BA} = EI\theta_B$ (columns, EI/4 stiffness); $M_{BC} = -60 + (2 \times 2EI/6)(2\theta_B + \theta_C) = -60 + (2EI/3)\,\theta_B$.
- Joint B: $EI\theta_B\,(1 + 2/3) = 60 \rightarrow EI\theta_B = 36 \rightarrow$ **$M_{BA} = 36$, $M_{AB} = 18$, $M_{BC} = -36$ kN·m** .
- **Stiffer beam → larger share of the load moment attracted by the beam; column moments smaller** (compare 45 earlier).

Example 9 — Hinged-base portal frame (symmetric UDL)

- Columns hinged at A and D → modified: $M_{BA} = (3EI/4)\theta_B$ ($F'=0$). $M_{BC} = -60 + (EI/3)\theta_B$.
- Joint B: $EI\theta_B\,(0.75 + 0.333) = 60 \rightarrow EI\theta_B = 55.38 \rightarrow$ **$M_{BA} = 41.54$ kN·m** , $M_{BC} = -41.54$ kN·m; $M_{AB} = 0$.
- Horizontal thrust $H = 41.54/4 =$ **10.38 kN**; beam centre moment = 90 − 41.54 = **48.46 kN·m** . Compared with the fixed-base frame: smaller thrust and column moment, but **larger sagging moment in the beam** .

9A. More Solved Examples — Beams

Example 10 — Propped cantilever (A fixed, B roller) with UDL

- $\theta_A = 0$, $M_{BA} = 0 \rightarrow$ use the modified equation for end A: $M_{AB} = F_{AB} - \tfrac{1}{2}F_{BA} + (3EI/L)(\theta_A - \psi)$.
- $M_{AB} = -wL^2/12 - \tfrac{1}{2}\,(wL^2/12) + 0 =$ **$-wL^2/8$** (hogging at the fixed end) √ — result in one step.
- $R_A = 5wL/8$, $R_B = 3wL/8$.

Example 11 — Two spans with different spans and different EI

Problem

$AB = 6$ m (EI), $BC = 4$ m ($2EI$); UDL 20 kN/m on both spans; A and C simply supported. Find M_B.

- $F'_{BA} = wL^2/8 = 20 \times 36/8 = 90$; $F'_{BC} = -20 \times 16/8 = -40$.
- $M_{BA} = 90 + (3EI/6)\theta_B = 90 + 0.5\,EI\theta_B$
- $M_{BC} = -40 + (3 \times 2EI/4)\theta_B = -40 + 1.5\,EI\theta_B$
- Joint B: $50 + 2\,EI\theta_B = 0 \rightarrow EI\theta_B = -25$.

Result

$M_{BA} = 90 - 12.5 = 77.5$ kN·m ; $M_{BC} = -40 - 37.5 = -77.5$ kN·m → $M_B = 77.5$ kN·m hogging.

- $R_A = 60 - 77.5/6 =$ **47.08 kN**; $R_C = 40 - 77.5/4 =$ **20.63 kN**; $R_B = 200 - 67.71 =$ **132.29 kN** .
- Observation: the **stiffer span attracts more moment** , as stiffness $= 3EI/L$ is larger for BC.

Example 12 — Three equal spans with central point loads

- Spans L, central load P in each span, ends pinned; $\theta_C = -\theta_B$ (symmetry).
- $F'_{BA} = PL/8 + \tfrac{1}{2}(PL/8) = 3PL/16$; $M_{BA} = 3PL/16 + (3EI/L)\theta_B$.
- $M_{BC} = -PL/8 + (2EI/L)(2\theta_B + \theta_C) = -PL/8 + (2EI/L)\theta_B$.
- Joint B: $PL/16 + 5EI\theta_B/L = 0 \rightarrow EI\theta_B = -PL^2/80$.

Result

$M_B = M_C = 3PL/20$ — for $L = 6$ m and $P = 40$ kN: **36 kN·m (hogging)** ; $R_A = P/2 - M_B/L = 20 - 6 =$ **14 kN** $= R_D$; $R_B = R_C = (3P - 28)/2 =$ **46 kN** (check: $14 + 46 + 46 + 14 = 120 = 3P$ √).

Example 13 — Joint carrying an external moment

- If a couple M_0 (clockwise) acts at joint B, the equilibrium equation becomes **$M_{BA} + M_{BC} = M_0$** (the joint moment is balanced by the member moments).

9B. Portal Frame — Extra Example and BMD

Example 14 — Fixed-base portal with a central point load on the beam

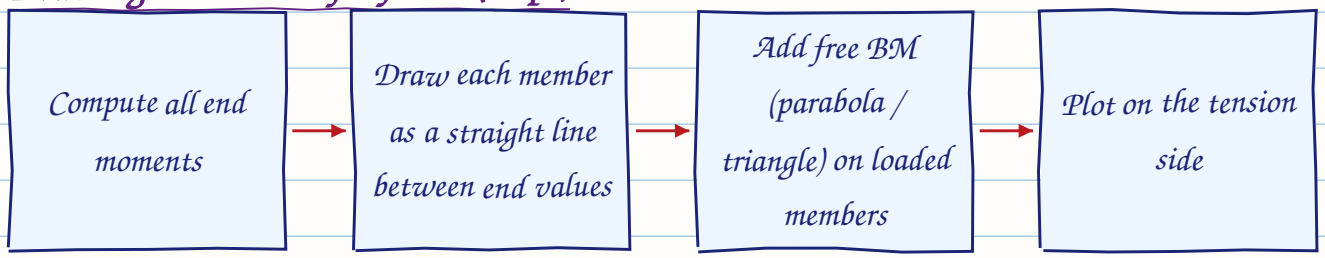
Problem

$h = 4$ m, $L = 6$ m, EI same; $P = 48$ kN at the centre of BC. A and D fixed.

- $F_{BC} = -PL/8 = -36$. By symmetry $\theta_C = -\theta_B$.
- $M_{BA} = EI\theta_B$; $M_{BC} = -36 + (EI/3)\theta_B$. Joint B: $(4/3)\,EI\theta_B = 36 \rightarrow$ **$EI\theta_B = 27$** .
- **$M_{BA} = 27$, $M_{AB} = 13.5$, $M_{BC} = -27$ kN·m** . Beam centre: $PL/4 - 27 = 72 - 27 =$ **+ 45 kN·m** .

- Horizontal reaction $H = (13.5 + 27)/4 =$ **10.13 kN**; $V_A = V_D =$ **24 kN**.

Drawing the BMD of a frame (steps)



- Plot the BM on the **tension side** of each member; at a rigid joint the BM is **continuous** in value — the moment going into the joint from one member equals the moment leaving by the other (joint equilibrium).
- Check: sum of moments at each joint = 0 and the horizontal reactions at the bases are equal and opposite (no sway).

Frame result	Fixed bases	Hinged bases	Remark
Column moment at base	22.5	0	hinged base has no moment
Moment at joint B	45	41.5	for UDL 20 kN/m, $h = 4$, $L = 6$
Beam centre BM	45	48.5	beam works harder when bases are hinged
Horizontal reaction	16.9	10.4	thrust at bases

9C. Fixed Beams and Propped Cantilever by Slope-Deflection (Quick Methods)

- **Fixed beam:** $\theta_A = \theta_B = 0$ and no settlement → end moments = fixed end moments F_{AB}, F_{BA} directly.
- **Fixed beam with settlement Δ:** $M_{AB} = F_{AB} - 6EI\Delta/L^2$, $M_{BA} = F_{BA} - 6EI\Delta/L^2$ (B sinks).
- **Support rotation θ_A:** $M_{AB} = F_{AB} + 4EI\theta_A/L$, $M_{BA} = F_{BA} + 2EI\theta_A/L$.
- **Propped cantilever with point load P at the centre:** $M_{AB} = F_{AB} - \frac{1}{2}F_{BA} = -PL/8 - PL/16 =$ **$-3PL/16$** √.

9D. Checks and Common Mistakes

No.	Point to check
1	Clockwise = positive for moments, θ and ψ — be consistent throughout.
2	FEM of a downward load: **− at the left end, + at the right end** of the span.
3	Use the **modified** equation only when the far end is hinged/roller and write $F - \frac{1}{2}F'$ correctly.

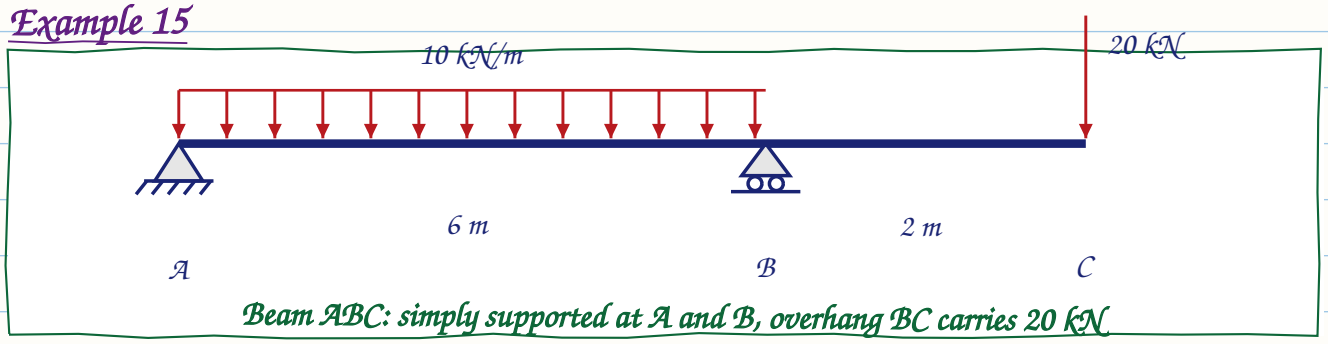
No.	Point to check
4	ψ is the same for both ends of the member; sign by a sketch.
5	Joint equation: ΣM (member end moments at the joint) = 0, not the sum of rotations.
6	The moment at an overhang end is known — do not treat it as unknown.
7	Verify by a second method (three-moment equation or ΣV = total load).
8	Cantilever overhang members have **zero stiffness** ; do not include them in $EI\theta$ terms.

9E. Practice Problems with Answers

No.	Problem	Answer
1	Two equal spans 5 m, UDL 24 kN/m, A and C pinned	$M_B = wL^2/8 = 75$ kN·m
2	Propped cantilever $L = 4$ m, UDL 15 kN/m (fixed at A)	$M_A = wL^2/8 = 30$ kN·m
3	Three equal spans 6 m, UDL 10 kN/m, ends pinned	$M_B = M_C = wL^2/10 = 36$ kN·m
4	Two equal spans 6 m, central load 60 kN in each	$M_B = 3PL/16 = 67.5$ kN·m
5	Fixed-base portal $h = 4$ m, $L = 6$ m, UDL 30 kN/m (EI same)	$M_{BA} = 67.5$; $M_{AB} = 33.75$ kN·m
6	Two equal spans 6 m, UDL 20 kN/m, B settles 10 mm, $EI = 30000$	$M_B = 65$ kN·m

9F. Beam with Overhang — Solved Example

Example 15



Beam ABC: simply supported at A and B, overhang BC carries 20 kN

- Overhang moment: $M_{BC} = -20 \times 2 = -40$ kN·m (known; the overhang is a cantilever, no stiffness).
- Joint B: $M_{BA} + M_{BC} = 0 \rightarrow M_{BA} = +40$ kN·m .
- Check the unknown rotation: $M_{BA} = F'_{BA} + (3EI/6)\theta_B = 45 + 0.5\,EI\theta_B = 40 \rightarrow EI\theta_B = -10$ (anticlockwise).
- Reactions: $R_A = 30 - 40/6 =$ **23.33 kN**; $R_B = (30 + 6.67) + 20 =$ **56.67 kN**; check: $23.33 + 56.67 = 80 = 60 + 20$ √.
- Max sagging in AB at $x = 23.33/10 = 2.33$ m: $M = 23.33 \times 2.33 - 5 \times 2.33^2 =$ **27.2 kN·m** .

9G. Shear Force and Bending Moment Values for Example 1

Section	Shear force (kN)	Bending moment (kN·m)
A	+ 49	0
2.45 m from A	0	+ 60.0 (max sagging)
B (left)	− 71	− 66
B (right)	+ 36.5	− 66
under 40 kN load (2 m from B)	+ 36.5 → − 3.5	+ 7.0
C	− 3.5	0

- Check: area of SFD between B and the point of zero moment gives BM change; BM is maximum where SF changes sign.

9H. Stiffness and Carry-Over — Quick Derivation

- Apply a unit rotation $\theta_A = 1$ at A while B is **fixed** ($\theta_B = 0$, $\psi = 0$): $M_{AB} = 4EI/L$ (stiffness) and $M_{BA} = 2EI/L \rightarrow$ **carry-over factor = ½** .
- If B is **hinged** : $M_{BA} = 0 \rightarrow$ from the two equations $2\theta_B + \theta_A = 0 \rightarrow \theta_B = -\theta_A/2 \rightarrow M_{AB} = (2EI/L)(2\theta_A - \theta_A/2) = 3EI\,\theta_A / L$.
- Relative displacement: $M_{AB} = M_{BA} = -6EI\psi/L \rightarrow$ settlement Δ gives $6EI\Delta/L^2$.
- These three results ($4EI/L$, $3EI/L$, $6EI/L^2$) are the entries of the **member stiffness matrix** — the slope-deflection method is the hand form of the **matrix stiffness method** .

Quantity	Far end fixed	Far end hinged
Rotation stiffness at near end	$4EI/L$	$3EI/L$
Carry-over moment	½ M	0
Moment for settlement Δ (per unit Δ)	$-6EI/L^2$ (both ends)	$-3EI/L^2$ (near end only)

9I. Choosing a Method — Quick Guide

Count D_s and D_k → Few redundants (D_s small)? Use force methods → Few joint displacements (D_k small)? Use slope-deflection / moment distribution

- **Fixed beam:** $D_s = 3$, $D_k = 0 \rightarrow$ slope-deflection is immediate (no unknowns).
- **Two-span beam with simple ends:** $D_k = 1$ (θ_B) → one equation.
- **Portal frame without sway (symmetric):** $D_k = 1$ effectively.

10. Frames with Unsymmetrical Loading (Notes)

- If the frame is unsymmetrical in loading but sway is prevented, **θ_B and θ_C are both unknown**; write two joint equations, solve simultaneously.
- If the frame **sways**, an extra unknown Δ (ψ for columns) appears, and an additional equation is needed — the **horizontal force (storey shear) equation** $\Sigma H = 0$. This is outside the present scope.
- Typical example: a portal with a horizontal load at beam level sways → solve with sway.

11. Relative Stiffness and Distribution (link to Unit V)

- Joint equilibrium shows that a joint moment M is shared in proportion to **stiffness $4EI/L$ (or $3EI/L$)**: this idea leads to **moment distribution** (Unit V) — a stepwise (iterative) form of slope deflection.
- Distribution factor DF = stiffness of member ÷ sum of stiffnesses at the joint.

12. Advantages and Limitations

Aspect	Advantages	Limitations
Method	systematic; works for beams and frames	simultaneous equations if many joints
Unknowns	only joint displacements (D_k)	large number for tall frames
Settlement	easy to include (ψ)	sign errors are common
Computer use	forms the basis of the matrix stiffness method	hand work lengthy for sway frames

13. Slope-Deflection vs Moment Distribution vs Three-Moment

Basis	Slope-deflection	Three-moment	Moment distribution
Unknowns	θ, ψ	support moments	none (iterative)
Equations	joint equilibrium	one per pair of spans	none
Best for	beams and frames	continuous beams	beams and frames
Result	exact	exact	exact (after convergence)

14. Quick Revision — Unit IV

Formulae

$M_{AB} = F_{AB} + (2EI/L)(2\theta_A + \theta_B - 3\psi)$ | $M_{BA} = F_{BA} + (2EI/L)(2\theta_B + \theta_A - 3\psi)$ | Far end hinged: $M_{AB} = F_{AB} - \tfrac{1}{2}F_{BA} + (3EI/L)(\theta_A - \psi)$ | $\psi = \Delta/L$ (clockwise +ve) | FEM: $\mp wL^2/12$, $\mp PL/8$, $\mp Pab^2/L^2$, $\pm Pa^2b/L^2$ | Joint: $\Sigma M = 0$.

Possible exam questions

1. Derive the slope-deflection equations for a member with end moments.
2. Analyse a two-span continuous beam by the slope-deflection method; draw BMD.
3. Analyse the continuous beam when one support settles.
4. Analyse a single-bay portal frame without sway using the slope-deflection method.
5. Explain the modified slope-deflection equation for a hinged far end.